

FLUTTER ☐ ☐ ☐ ☐

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2-4 /



Flutter

1. Widget Element RenderObject
2. setState
3. const
4. GlobalKey



- 5.
6. 2MB OOM
7. Future microtask isolate
- 8.



9. Riverpod/Bloc/GetX
10. App
- 11.
- 12.



13. WebSocket
- 14.
15. WebSocket



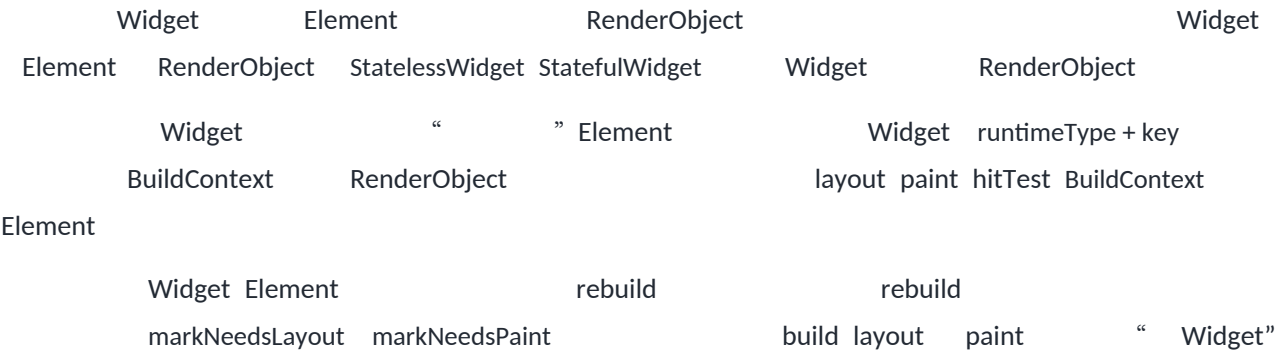
- 16.
17. MethodChannel EventChannel
18. Android/iOS
- 19.



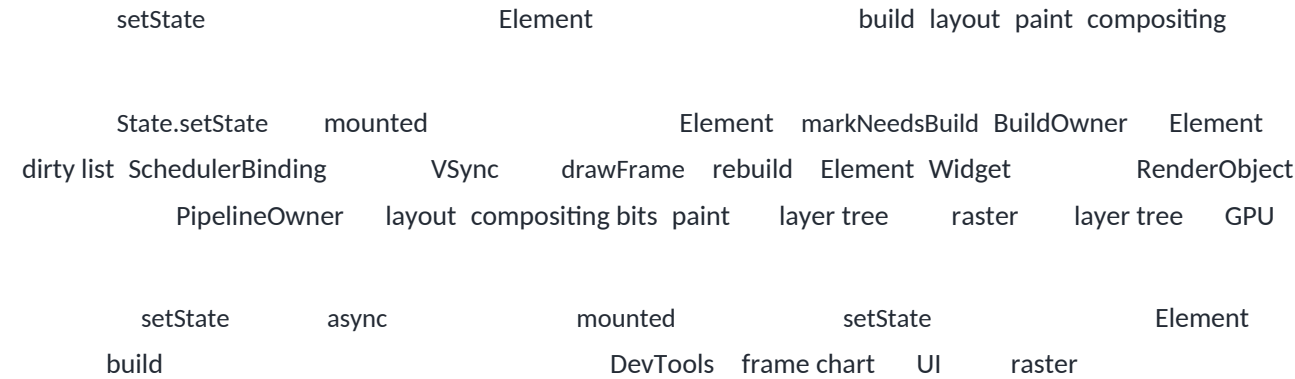
- 20.



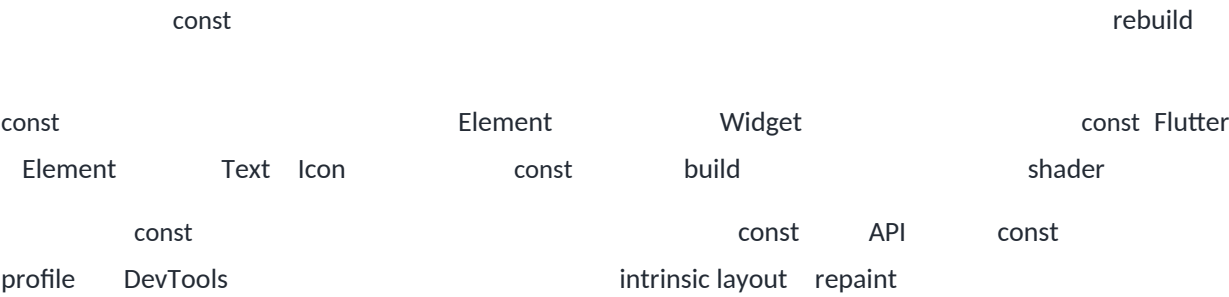
1. Widget Element RenderObject



2. setState



3. const



4. GlobalKey



FormState.validate() ScaffoldMessenger State
reparent deactivate/activate InheritedWidget GlobalKey key
build GlobalKey
GlobalKey State “ ” ValueKey(id) Controller
LayoutBuilder GlobalKey

□ □ □ □ □ □ □

5.

60Hz 16.7ms 120Hz 8.3ms UI build/layout raster
shader
profile DevTools Performance frame chart UI jank raster jank rebuild stats Track
Widget Builds Repaint Rainbow CPU profiler memory/image cache UI builder
item build JSON/ / IO shrinkWrap IntrinsicHeight Raster
saveLayer repaint
ListView.builder/Sliver item key build repaint
RepaintBoundary P90/P99 frame time “ ”

6. 2MB OOM

2MB × × 4 bytes CPU GPU
8000×6000 JPEG RGBA 183MB GPU EXIF
ImageCache Dart heap Dart heap
DPR cacheWidth/cacheHeight ResizeImage “ 100px
4000px” DevTools Memory Android dumsys
meminfo/Profiler iOS Instruments Dart heap native heap GPU OOM ImageCache

7.Future microtask isolate

Future microtask isolate event queue isolate
Future
async/await isolate microtask queue event timer IO microtask
microtask UI IO Future()/Timer event queue IO
await UI isolate

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CPU JSON UI isolate Isolate.run worker isolate isolate
TransferableTypedData IO Future microtask CPU
isolate

8.

GC native
20 profile Dart heap external/native memory RSS DevTools
Memory diff snapshot retained size retaining path StreamSubscription/Timer
AnimationController/ScrollController dispose State Provider EventChannel
observer/delegate
Dart heap RSS Android Studio Profiler LeakCanary dumpsys meminfo iOS Instruments
Leaks/Allocations bitmap GC retained objects
retaining path



9. Riverpod/Bloc/GetX

Riverpod Bloc
GetX
Riverpod BuildContext family autoDispose override Bloc event-
state GetX API
spike Hello World domain/use case
loading/error/data dispose “ ”

10. App

feature-first presentation
application/domain data
Presentation UI state application/use case domain data API DB DTO
public API data core core DI
DTO UI
feature package lint API feature flag use case
repository ADR

11.

		cursor		id		
Offset		nextCursor		createdAt desc, id desc	itemsByld +	
orderedIds	id upsert	append		initial refresh	loadMore end	
error	cursor					
	generation requestld			generation	debounce	
switchLatest		id				

12.

	entityld				
FavoriteRepository/Store	Map<id, FavoriteState>		store	id	
operationld					
PUT/DELETE	toggle		store	/	
WebSocket/	UI				



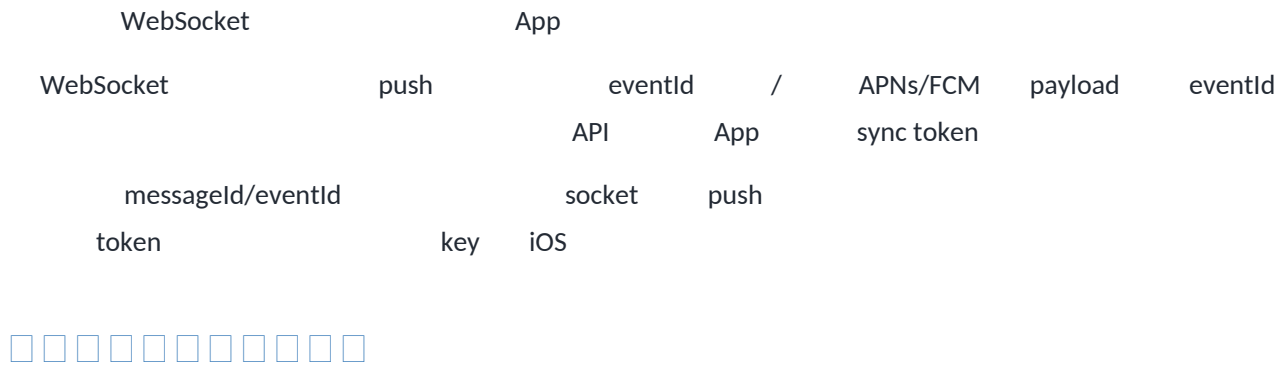
13. WebSocket

WebSocket	“ ”	id			
channel/conversationld	messageld	serverSeq	serverTime	payload	messageld
lastAckSeq	resume token	lastSeq	seq	expectedSeq	/
API	expectedSeq		ack		
clientMessageld	pending	sent	acknowledged	failed	id
WebSocket REST		serverSeq			

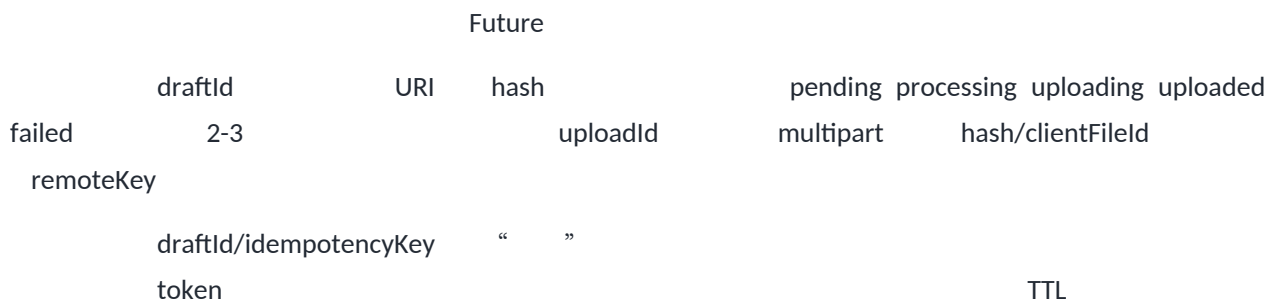
14.

					TCP
ping	pong	25	ping 10	pong	App
Android/iOS					
full jitter	30-60	401	token		connectivity
generation	socket			ready	

15. WebSocket



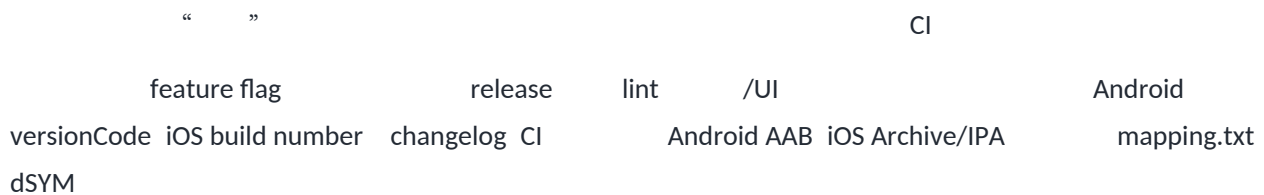
16.



17. MethodChannel EventChannel



18. Android/iOS



Android Play Console / Data safety release notes staged rollout iOS App Store
Connect TestFlight / phased release crash-free
users ANR / tag

19.

feature flag
1% 5% 20% / crash-free
ANR OOM P95/P99 trace breadcrumb
release id
flag
kill switch expand-migrate-contract “ ”



20.

STAR										
Situation	Android		3-5		OOM		Crash	native allocation failed		Dart
	crash-free users		99.7%	98.9%						
Task										
Action	DevTools	Dart heap		RSS	Graphics		GPU	dumpsys	meminfo	Android
Profiler		120dp		4000px		ImageProvider		MB		
	MB	CDN	URL	xDPR						
A/B	PSS	GPU	P99 frame time							
Result	20	PSS	620MB	260MB	P99	46ms	19ms	crash-free		99.75%
	Dart/native/GPU									



15 60-120 30-60 20
“ ” “ ” “ ”